Waqas Ali

Hong Kong

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Summary

Software engineer and tech lead working on data infrastructure and distributed systems in finance.

Experience

- **Qube Research & Technologies** — Hong Kong
  Software Engineer (Quantitative Technologist); Tech Lead, DataOps and Automation — *April 2024 – Present*
  - Tech lead on the firm's core data platform, owning the data lifecycle and catalog. Re-architected its storage so data can live in several databases at once, and designed the orchestration that migrates and syncs data across them. I also lead developer experience for the larger team. Rust, ClickHouse, PostgreSQL, Kafka and Kubernetes, across AWS and on-premise.

- **Credit Suisse** — Hong Kong
  Software Engineer; Assistant Vice President — *August 2020 – March 2024*
  - **Wealth Management Analytics & Lead Management:** Rebuilt the analytics platform achieving 36x improvement in query performance (from 1m 48s to 3s for the same query). Implemented a lead management system for capturing leads and tracking conversions (4500 leads and USD 50M tracked orders in first 6 months). Enabled data-driven decision making through performance analysis of investment ideas and client facing teams. Used Prefect for data ingestion workflows, ClickHouse/SQL for OLAP (Online Analytical Processing) and Python for the REST API.
  - **Instrument Events:** Proactively notified clients of significant events such as research rating downgrades, bond maturities, and price fluctuations, by monitoring data for 50,000 instruments on a daily basis. Analyzed 1M rows under 10 minutes using Clojure and Apache Spark. Achieved auto-scaling and cost optimization by migrating workflows from a self-maintained Spark cluster to the company's Kubernetes cluster, simplifying infrastructure and enabling on-demand resource usage.
  - **Risk Data Aggregation:** Doubled throughput and cut wasted compute by moving a pipeline from threads in a monolith on a virtual machine to 25 task processors on Kubernetes, autoscaling on queue size. (Kotlin, Kafka, ZooKeeper, Kafka Connect, Flux GitOps)
  - **Market Data:** Cut production errors by 80% by rewriting an unreliable closed-source JNI library in Clojure, implementing the PermServer protocol over TCP sockets with core.async.
  - **Observability:** Implemented the OpenTelemetry collector, ClickHouse for logs and metrics, Kafka Connect and Prometheus Alertmanager. Wrote alerts as infrastructure-as-code with unit tests, and was on call for the platform.

Education

- **The University of Hong Kong**
  Bachelor of Engineering in Computer Science (Minor in Cognitive Science) — *3.51/4.30 GPA*
  Full Scholarship — *Sept. 2016 – May 2020*

Projects

- **studentbase.app:** Developed a platform for high school students used by 25,000 unique users every month. Implemented features such as content upload, moderation, comments, upvotes, past paper solving, and mistake tracking. Seed-funded with HK$100,000 by Hong Kong Science Park; led four part-time engineers and an intern. Leveraged React, Next.js, TypeScript, Node.js, GraphQL, Prisma, PostgreSQL, Redis, Prometheus, GitHub Actions for CI/CD while utilizing Terraform for provisioning/configuration of DNS, Certificates, Virtual Machines, S3, and CDN.

Open Source Contributions

- **Geni:** Added the ability to use nested Clojure data structures with Apache Spark. Implemented transformation to Spark DataFrames with automatic type inference. #336 #340

Skills

- **Programming Languages:** Rust, Clojure, Python, Kotlin, JavaScript/TypeScript, SQL
- **Data Engineering:** Apache Spark, Kafka, Prefect, ClickHouse, PostgreSQL, Arrow, Parquet, S3
- **Web Development:** HTML, CSS, React, Next.js, Node.js, GraphQL, MongoDB
- **Infrastructure:** Linux, Kubernetes, AWS, Docker, Terraform, OpenTelemetry, Prometheus, Bazel, GitLab CI